

# Report 16-06-2026

**Abstract—**

**Index Terms—**

## I. INTRODUCTION

Joint Radar-Communication (JRC) systems have emerged as a promising technology to address spectrum congestion and reduce hardware cost, size, and power consumption by sharing a single waveform for both communication and sensing tasks. Phase-Modulated Continuous Wave (PMCW) signaling is a particularly attractive candidate for JRC due to its constant envelope property, high processing gain, and resilience against interference. By modulating a carrier with pseudo-random binary sequences (PRBS), PMCW systems can achieve excellent radar resolution through matched filtering while conveying digital data.

This paper presents a detailed study of the PMCW JRC pipeline. We formulate the signal generation process and define the frame structure, which incorporates both sounding preambles and data blocks. We then introduce the channel propagation models for both the downlink communication path and the round-trip radar backscattering path. Subsequently, we detail the step-by-step algorithms for communication data demodulation at the UE receiver and target parameter extraction (range, velocity, and angle) at the BS radar receiver.

**Notations:** Subscripts:

- “sound” is related to channel estimation;
- “sense” is related to the sensing signal;
- “comm” is related to the communication data signal at UE;
- “data” is related to the communication data signal at BS.

**In this report, I present 3 main points:**

- Mathematics baseline of PMCW, presented in Section II,
- System models, presented in Section III,
- Proposed problem, presented in Section IV.

## II. MATHEMATICS BASELINE

### A. Waveform Generation and Modulation

A PMCW JRC system transmits a continuous-wave carrier that is phase-modulated by a PRBS, typically a maximum-length sequence (m-sequence).

1) *Spreading Code Definition:* Let  $c[n] \in \{+1, -1\}$  denote an m-sequence of length  $N_{\text{chip}} = 2^m - 1$  chips, where  $n = 0, 1, \dots, N_{\text{chip}} - 1$  and  $m$  is the shift register length. Given a chip duration of  $T_c$ , the modulation bandwidth is approximated as:

$$B \approx \frac{1}{T_c}. \quad (1)$$

The continuous-time code waveform  $c(t)$  is generated by modulating the sequence onto a chip-shaping pulse  $p(t)$ . Assuming a rectangular pulse of duration  $T_c$ , we express the continuous-time code waveform as:

$$c(t) = \sum_{n=0}^{N_{\text{chip}}-1} c[n]p(t - nT_c), \quad (2)$$

where

$$p(t) = \begin{cases} 1, & 0 \leq t < T_c \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

The duration of a single complete m-sequence block is given by:

$$T_m = N_{\text{chip}}T_c. \quad (4)$$

The generated PRBS sequence waveform  $c(t)$  is illustrated in Fig. 1.

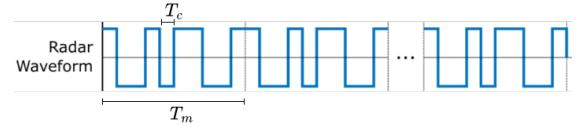


Fig. 1: Continuous-time PRBS code waveform  $c(t)$ .

### 2) Joint Radar-Communication (JRC) Frame Structure:

To transmit communication data while maintaining radar sensing capabilities, the transmission frame is divided into  $M$  blocks, indexed by the slow-time variable  $m = 0, 1, \dots, M - 1$ . Each block consists of two distinct periods:

- 1) **Sounding/Preamble Period:** A pure, unmodulated PRBS sequence dedicated to communication channel estimation at the UE, which is essential for subsequent data demodulation.
- 2) **Data Period:** The PRBS sequence modulated by a Binary Phase-Shift Keying (BPSK) data symbol  $b_m \in \{+1, -1\}$ .

The total duration of one JRC block is  $T_{\text{block}} = 2T_m$ . Consequently, the continuous-time baseband transmitted signal for block  $m$  is:

$$s_m(t) = c(t - mT_{\text{block}}) + b_m c(t - mT_{\text{block}} - T_m), \quad (5)$$

where the first term represents the sounding signal  $s_{\text{sound}}(t)$  and the second term represents the communication signal  $s_{\text{comm},m}(t)$ , which is the product of the data symbol  $b_m$  and the spreading sequence. The JRC block structure and the modulation pipeline are illustrated in Fig. 2 and Fig. 3, respectively.

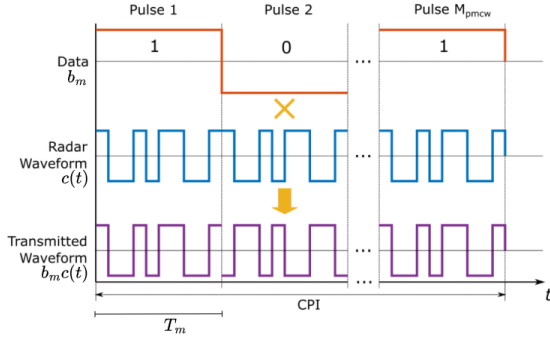


Fig. 2: PMCW JRC transmitter block diagram showing sounding and data modulation paths.

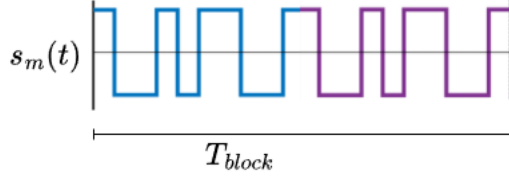


Fig. 3: JRC block structure displaying the sounding preamble and the data period.

### B. Signal Propagation and Channel Model

1) *Downlink Communication Channel (Base Station  $\rightarrow$  UE:  $h_{\text{sound},m}(t) \approx h_{\text{comm},m}(t)$ ):* Let  $h_{\text{comm},m}$  represent the channel coefficient,  $\tau_{\text{comm}}$  represent the propagation delay, and  $f_{d,\text{comm}}$  denote the communication Doppler shift. The received baseband signal at the UE during block  $m$  is expressed as:

$$r_{\text{comm},m}(t) = h_{\text{comm},m}(t) * s_m(t) e^{j2\pi f_{d,\text{comm}} m T_{\text{block}}} + w_{\text{comm},m}(t), \quad (6)$$

where  $*$  denotes the continuous-time convolution operator,  $h_m(t)$  is the channel impulse response,  $f_d$  is the Doppler frequency shift, and  $w_m(t)$  represents Additive White Gaussian Noise (AWGN). The received signal  $r_m(t)$  consists of two periods: a sounding period  $r_{\text{sound},m}(t)$  (serves for the channel estimation purpose) and a communication/sensing period  $r_{\text{comm},m}(t)$ . Correspondingly, the channel impulse response  $h_m(t)$  can be partitioned into  $h_{\text{sound},m}(t)$  and  $h_{\text{comm},m}(t)$ . Under the assumption of a slow-varying channel, the channel characteristics during the communication period are approximated by those estimated during the sounding period, yielding:

$$h_{\text{sound},m}(t) \approx h_{\text{comm},m}(t). \quad (7)$$

The slow-time Doppler phase term,  $e^{j2\pi f_{d,\text{comm}} m T_{\text{block}}}$ , accounts for the phase shift that accumulates from one block to the next. Because the block duration  $T_{\text{block}}$  is relatively large, the block-to-block phase shift is substantial. This accumulation over the CPI creates a phase oscillation that is subsequently resolved by the Doppler FFT to determine the velocity.

Substituting the transmitted signal  $s_m(t)$  into the above model, we obtain:

$$r_{\text{comm},m}(t) = h_{\text{comm},m}(t) * \left[ c(t - mT_{\text{block}}) + b_m c(t - mT_{\text{block}} - T_m) \right] e^{j2\pi f_{d,\text{comm}} m T_{\text{block}}} + w_{\text{comm},m}(t). \quad (8)$$

2) *Radar Sensing Channel (Base Station  $\rightarrow$  Target  $\rightarrow$  Base Station:  $h_{\text{sound},m}(t)$ ):* We consider a target located at a range  $R$ , moving with a radial velocity  $v$  at an azimuth angle of arrival  $\theta$ . The round-trip delay  $\tau$  and the target-induced Doppler frequency shift  $f_d$  are respectively defined as:

$$\tau = \frac{2R}{c_0}, \quad f_d = \frac{2v}{\lambda} = \frac{2vf_c}{c_0}, \quad (9)$$

where  $c_0$  is the speed of light and  $\lambda$  is the carrier wavelength.

For a colocated MIMO radar with Uniform Linear Arrays (ULAs) at both transmitter and receiver, let  $u$  ( $u = 0, 1, \dots, N_{\text{tx}} - 1$ ) denote the transmitter antenna index and  $v_r$  ( $v_r = 0, 1, \dots, N_{\text{rx}} - 1$ ) denote the receiver antenna index. The received baseband signal at receiver antenna  $v_r$  corresponding to the transmission from antenna  $u$  during block  $m$  is modeled as:

$$y_{v_r,u,m}(t) = h_{\text{sense},m}(t) * s_m(t) e^{ju\pi \sin \theta} e^{jv_r N_{\text{tx}} \pi \sin \theta} \times e^{j2\pi f_d m T_{\text{block}}} + w_{v_r,u,m}(t), \quad (10)$$

where  $w_{v_r,u,m}(t)$  is the additive noise.

By defining a virtual antenna index<sup>1</sup>  $k = u + v_r N_{\text{tx}}$  ( $k = 0, 1, \dots, N_{\text{tx}} N_{\text{rx}} - 1$ ), we can synthesize a virtual ULA. The received signal at the virtual element  $k$  is then expressed as:

$$y_{k,m}(t) = h_{\text{sense},m}(t) * \left[ c(t - mT_{\text{block}}) + b_m c(t - mT_{\text{block}} - T_m) \right] e^{jk\pi \sin \theta} e^{j2\pi f_d m T_{\text{block}}} + w_{k,m}(t). \quad (11)$$

### C. Communication Data Demodulation (at UE)

The UE samples the received baseband signal at the chip rate  $f_s = 1/T_c$ , resulting in  $2N_{\text{chip}}$  discrete-time samples per block. In this configuration, the sampling rate matches the PRBS chip rate, though they may differ in other system configurations. Let  $r_{\text{sound},m}[n]$  and  $r_{\text{data},m}[n]$  represent the sampled sequences during the sounding and data periods of block  $m$ , respectively. Here, the channel during the communication period is assumed to be quasi-static relative to the sounding period, i.e.,  $h_{\text{comm}}[n] \approx h_{\text{sound}}[n]$ . The communication data demodulation pipeline at the UE is depicted in Fig. 4.

<sup>1</sup>To better understand why we can set these virtual antennas, see examples in Appendix A.

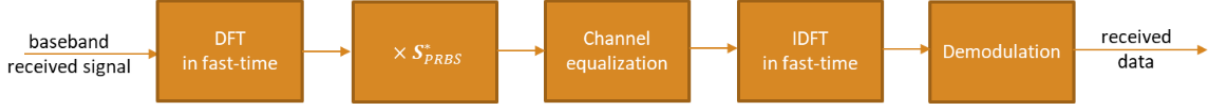


Fig. 4: Communication data demodulation pipeline at the UE receiver.

1) *Step 1: Channel Estimation (Sounding Preamble):* During the sounding period of block  $m$ , the received samples are given by:

$$r_{\text{sound},m}[n] = h_{\text{comm}}[n] * c[n] + w_{\text{sound},m}[n], \quad (12)$$

where  $c[n]$  represents the sampled PRBS sequence. Taking the  $N_{\text{chip}}$ -point DFT of both sides yields:

$$R_{\text{sound},m}[k_f] = H_{\text{comm}}[k_f]C[k_f] + W_{\text{sound},m}[k_f], \quad (13)$$

where  $k_f$  is the subcarrier index. Using the known preamble sequence  $C[k_f]$ , the frequency-domain channel estimate  $\hat{H}_{\text{comm}}[k_f]$  is computed via least-squares estimation as:

$$\hat{H}_{\text{comm}}[k_f] = \frac{R_{\text{sound},m}[k_f]}{C[k_f]} = H_{\text{comm}}[k_f] + \frac{W_{\text{sound},m}[k_f]}{C[k_f]}. \quad (14)$$

2) *Step 2: DFT of the Received Data Period:* During the data period, the received discrete-time signal is:

$$r_{\text{comm},m}[n] = b_m(h_{\text{comm}}[n] * c[n]) + w_{\text{comm},m}[n]. \quad (15)$$

Applying the  $N_{\text{chip}}$ -point DFT, we obtain:

$$R_{\text{comm},m}[k_f] = b_m H_{\text{comm}}[k_f]C[k_f] + W_{\text{comm},m}[k_f]. \quad (16)$$

3) *Step 3: Despreading (Matched Filtering):* To perform despreading, we multiply the received data spectrum by the complex conjugate of the reference PRBS spectrum  $C^*[k_f]$ , yielding:

$$\begin{aligned} Z_{\text{mf},m}[k_f] &= R_{\text{comm},m}[k_f]C^*[k_f] \\ &= b_m H_{\text{comm}}[k_f]|C[k_f]|^2 + W_{\text{comm},m}[k_f]C^*[k_f]. \end{aligned} \quad (17)$$

Due to the flat power spectral density of m-sequences, the power spectrum satisfies  $|C[k_f]|^2 \approx N_{\text{chip}}$  for all  $k_f$ .

In the time domain, this multiplication corresponds to circular cross-correlation. The resulting output in the time domain exhibits a single sharp peak at the true path delay, while other bins remain close to zero (i.e., in the form of  $[0, 0, \dots, \text{Peak}, \dots, 0]$ ). Crucially, the matched filtering does not demodulate the signal back to the raw block of +1 and -1 chips, but rather compresses the energy of the spreading sequence into a single peak.

4) *Step 4: Frequency Domain Equalization:* To compensate for channel-induced amplitude and phase variations, the matched-filtered spectrum is equalized by dividing it by the estimated channel response:

$$\begin{aligned} Z_{\text{eq},m}[k_f] &= \frac{Z_{\text{mf},m}[k_f]}{\hat{H}_{\text{comm}}[k_f]} \\ &\approx b_m |C[k_f]|^2 + \tilde{W}_m[k_f] \\ &\approx b_m N_{\text{chip}} + \tilde{W}_m[k_f], \end{aligned} \quad (18)$$

where  $\tilde{W}_m[k_f]$  represents the filtered noise.

5) *Step 5: IDFT to Time Domain:* Applying the IDFT transforms the equalized signal back to the time domain, producing a sharp autocorrelation peak at the path delay bin  $d$ :

$$z_{\text{eq},m}[n] = \text{IDFT}_{k_f}\{Z_{\text{eq},m}[k_f]\} = b_m N_{\text{chip}} \delta[n - d] + \tilde{w}_m[n]. \quad (19)$$

6) *Step 6: Symbol Decision:* The estimate of the BPSK symbol  $\hat{b}_m$  is obtained by extracting the sign of the real part of the signal at the peak index  $d$ :

$$\hat{b}_m = \text{sign}(\text{Re}\{z_{\text{eq},m}[d]\}). \quad (20)$$

The decoded information bit is subsequently retrieved as:

$$\hat{d}_{\text{bit},m} = \frac{1 - \hat{b}_m}{2} \in \{0, 1\}. \quad (21)$$

#### D. Radar Sensing Demodulation (at Base Station)

The BS functions as the radar receiver, processing the reflected echoes using its pre-existing knowledge of both the spreading sequence  $c[n]$  and the transmitted BPSK symbols  $b_m$ . The overall radar sensing demodulation and parameter extraction pipeline at the BS is shown in Fig. 5.

1) *Step 1: Removing Communication Data:* During the data period, the received signal at virtual antenna  $k$  and block  $m$  contains the BPSK symbol  $b_m$  and is written as:

$$\begin{aligned} r_{\text{data},k,m}[n] &= b_m h_{\text{sense},k,m}[n] * c[n - n_\tau] e^{jk\pi \sin \theta} \\ &\quad \times e^{j2\pi f_d m T_{\text{block}}} + w_{\text{data},k,m}[n]. \end{aligned} \quad (22)$$

Since the BS has full knowledge of the transmitted BPSK sequence  $b_m \in \{+1, -1\}$ , the communication modulation can be removed by multiplying the received signal by  $b_m$ . Recognizing that  $b_m^2 = 1$ , we obtain the stripped data signal:

$$\begin{aligned} r'_{\text{data},k,m}[n] &= r_{\text{data},k,m}[n] \cdot b_m \\ &= h_{\text{sense},k,m}[n] * c[n - n_\tau] e^{jk\pi \sin \theta} e^{j2\pi f_d m T_{\text{block}}} \\ &\quad + b_m w_{\text{data},k,m}[n]. \end{aligned} \quad (23)$$

2) *Step 2: Coherent Integration of Sounding and Data Periods:* To maximize the sensing SNR, the unmodulated sounding preamble is coherently integrated with the stripped data period. The received signal during the sounding period is given by:

$$\begin{aligned} r_{\text{sense},k,m}[n] &= h_{\text{sense},k,m}[n] * c[n - n_\tau] e^{jk\pi \sin \theta} \\ &\quad \times e^{j2\pi f_d m T_{\text{block}}} + w_{\text{sense},k,m}[n]. \end{aligned} \quad (24)$$

Summing the sounding signal and the stripped data signal yields the integrated signal:

$$\begin{aligned} r_{\text{int},k,m}[n] &= r_{\text{sense},k,m}[n] + r'_{\text{data},k,m}[n] \\ &= 2h_{\text{sense},k,m}[n] * c[n - n_\tau] e^{jk\pi \sin \theta} e^{j2\pi f_d m T_{\text{block}}} \\ &\quad + w'_{\text{int},k,m}[n], \end{aligned} \quad (25)$$



Fig. 5: Radar sensing demodulation and parameter extraction pipeline at the BS.

where the integrated noise is  $w'_{\text{int},k,m}[n] = w_{\text{sense},k,m}[n] + b_m w_{\text{data},k,m}[n]$ .

Assuming the noise terms  $w_{\text{sound},k,m}[n]$  and  $w_{\text{data},k,m}[n]$  are independent complex AWGN variables, each with variance  $\sigma^2$ , the variance of the integrated noise is analyzed as follows. For any complex random variable  $X$  and scalar  $a$ , the variance scales as  $\text{Var}(aX) = |a|^2 \text{Var}(X)$ . Since BPSK symbols satisfy  $|b_m|^2 = 1$ , we have:

$$\text{Var}(b_m w_{\text{data},k,m}) = |b_m|^2 \text{Var}(w_{\text{data},k,m}) = \sigma^2. \quad (26)$$

Thus, the variance of the integrated noise is:

$$\text{Var}(w'_{\text{int},k,m}) = \text{Var}(w_{\text{sound},k,m}) + \text{Var}(b_m w_{\text{data},k,m}) = 2\sigma^2. \quad (27)$$

As a result of this coherent integration:

- The signal amplitude doubles from  $A$  to  $2A$ , corresponding to a four-fold (+6 dB) increase in signal power.
- The noise variance doubles from  $\sigma^2$  to  $2\sigma^2$ , representing a two-fold (+3 dB) increase in noise power.
- The overall SNR becomes:

$$\text{SNR}_{\text{integrated}} = \frac{(2A)^2}{2\sigma^2} = 2 \times \frac{A^2}{\sigma^2} = 2 \times \text{SNR}_{\text{single\_period}}, \quad (28)$$

which yields a net +3 dB SNR improvement compared to utilizing only a single period.

3) *Step 3: Range Processing (Matched Filtering in Fast-Time)*: Range extraction is performed by executing circular cross-correlation in the frequency domain for computational efficiency. The procedure is as follows:

- 1) **Fast-Time DFT**: Transform the integrated sequence into the frequency domain:

$$R_{\text{int},k,m}[k_f] = \text{DFT}_n \{r_{\text{int},k,m}[n]\}. \quad (29)$$

- 2) **Conjugate Spectrum Multiplication**: Multiply by the complex conjugate of the reference PRBS spectrum to perform matched filtering:

$$Z_{\text{mf},k,m}[k_f] = R_{\text{int},k,m}[k_f] C^*[k_f]. \quad (30)$$

This is equivalent to circular cross-correlation in the time domain, which is performed to resolve the propagation delays (peaks). We execute this multiplication in the frequency domain for computational efficiency.

- 3) **Fast-Time IDFT**: Convert back to the time domain to resolve the delay profile:

$$\begin{aligned} \mathcal{X}[p, k, m] &= \text{IDFT}_{k_f} \{Z_{\text{mf},k,m}[k_f]\} \\ &= 2(h_{k,m}[p] * R_{cc}[p]) + \tilde{w}_{k,m}[p], \end{aligned} \quad (31)$$

where  $R_{cc}[p]$  is the circular autocorrelation function of the spreading sequence, and  $p$  represents the fast-time range bin.

Since the m-sequence exhibits near-ideal circular autocorrelation, we can approximate  $R_{cc}[p] \approx N_{\text{chip}} \delta[p]$ . Thus, the matched filter output simplifies to:

$$\mathcal{X}[p, k, m] \approx 2N_{\text{chip}} h_{k,m}[p] + \tilde{w}_{k,m}[p]. \quad (32)$$

By substituting a multi-target (multipath) channel model consisting of  $L$  discrete paths, the output is represented as:

$$\begin{aligned} \mathcal{X}[p, k, m] &\approx 2N_{\text{chip}} \sum_{l=1}^L \alpha_l \delta[p - n_{\tau,l}] e^{jk\pi \sin \theta_l} e^{j2\pi f_d l m T_{\text{block}}} \\ &\quad + \tilde{w}_{k,m}[p], \end{aligned} \quad (33)$$

where path  $l$  is characterized by a channel coefficient  $\alpha_l$ , a spatial phase progression term  $e^{jk\pi \sin \theta_l}$ , a Doppler phase term  $e^{j2\pi f_d l m T_{\text{block}}}$ , and a round-trip delay  $\tau_l$  (corresponding to the sampled bin  $n_{\tau,l} = \tau_l f_s$ ).

Assuming  $R_{cc}[p] \approx N_{\text{chip}}$  at  $p = n_{\tau}$  and is approximately zero elsewhere:

$$\begin{aligned} \mathcal{X}[p, k, m] &\approx \begin{cases} 2\alpha N_{\text{chip}} e^{jk\pi \sin \theta} e^{j2\pi f_d m T_{\text{block}}} + \tilde{w}_{k,m}[p], & p = n_{\tau} \\ \text{noise}, & p \neq n_{\tau}. \end{cases} \end{aligned} \quad (34)$$

The target range  $R$  is computed from the peak range bin  $p_{\text{peak}} = n_{\tau}$  using:

$$R = \frac{c_0 \tau_{\text{peak}}}{2} = \frac{c_0 p_{\text{peak}} T_c}{2} = \frac{c_0 p_{\text{peak}}}{2f_s}. \quad (35)$$

The corresponding range resolution is  $\Delta R = \frac{c_0}{2f_s}$ , and the maximum unambiguous range is  $R_{\text{max}} = \frac{c_0 T_m}{2}$ .

4) *Step 4: Velocity Processing (Doppler FFT in Slow-Time)*: At the identified peak range bin  $p_{\text{peak}}$ , the signal sequence across the slow-time blocks  $m$  is extracted as:

$$s_{\text{slow}}[m] = \mathcal{X}[p_{\text{peak}}, k, m] = 2\alpha N_{\text{chip}} e^{jk\pi \sin \theta} e^{j2\pi f_d m T_{\text{block}}}. \quad (36)$$

To estimate the Doppler frequency  $f_d$ , we apply a slow-time window function  $w_{\text{slow}}[m]$  (e.g., a Hann window) and perform an  $M$ -point DFT over the  $M$  accumulated blocks:

$$\mathcal{X}_{\text{RD}}[p_{\text{peak}}, k, q] = \sum_{m=0}^{M-1} s_{\text{slow}}[m] w_{\text{slow}}[m] e^{-j\frac{2\pi}{M} q m}. \quad (37)$$

This operation yields a peak in the range-Doppler map at the Doppler bin  $q_{\text{peak}}$ . After centering the spectrum (e.g., utilizing a circular shift operation), the Doppler frequency shift is determined as:

$$f_d = \frac{q_{\text{peak}} - M/2}{MT_{\text{block}}}. \quad (38)$$

The target's radial velocity  $v$  is then calculated by:

$$v = \frac{f_d \lambda}{2} = \frac{f_d c_0}{2f_c}. \quad (39)$$

The velocity resolution is  $\Delta v = \frac{\lambda}{2MT_{\text{block}}}$ , and the maximum unambiguous velocity is given by  $v_{\text{max}} = \pm \frac{\lambda}{4T_{\text{block}}}$ .

5) *Step 5: Spatial Processing (Angle of Arrival Estimation)*: –

**5.1 MIMO Array Spatial Coding:**<sup>2</sup>To resolve the target's azimuth angle of arrival (AoA)  $\theta$ , a colocated MIMO radar configuration is employed. By spacing the  $N_{\text{tx}}$  transmit antennas and  $N_{\text{rx}}$  receive antennas appropriately, we can synthesize a virtual array without spatial overlap:

- **Tx Antenna Spacing:**  $d_{\text{tx}} = \frac{\lambda}{2}$ , which defines the physical positions of transmit elements as  $x_{\text{tx}}[u] = u \frac{\lambda}{2}$  for  $u = 0, \dots, N_{\text{tx}} - 1$ .
- **Rx Antenna Spacing:**  $d_{\text{rx}} = N_{\text{tx}} \frac{\lambda}{2}$ , defining the physical positions of receive elements as  $x_{\text{rx}}[v_r] = v_r N_{\text{tx}} \frac{\lambda}{2}$  for  $v_r = 0, \dots, N_{\text{rx}} - 1$ .

The signal transmitted from antenna element  $u$  during block  $m$  is:

$$s_{u,m}(t) = s_m(t) e^{j u \frac{2\pi}{\lambda} d_{\text{tx}} \sin \theta} = s_m(t) e^{j u \pi \sin \theta}. \quad (40)$$

After reflecting off the target, the received baseband signal at receiver antenna  $v_r$  due to transmitter  $u$  during block  $m$  is:

$$y_{v_r,u,m}(t) = h_{\text{sense},m}(t) * s_m(t) e^{j u \pi \sin \theta} e^{j v_r N_{\text{tx}} \pi \sin \theta} \times e^{j 2\pi f_d m T_{\text{block}}} + w_{v_r,u,m}(t). \quad (41)$$

By indexing the virtual array elements as  $k = u + v_r N_{\text{tx}}$  for  $k = 0, \dots, N_{\text{virtual}} - 1$  (where  $N_{\text{virtual}} = N_{\text{tx}} N_{\text{rx}}$ ), the received signal at the virtual element  $k$  is simplified to:

$$y_{k,m}(t) = h_{\text{sense},m}(t) * \left[ c(t - mT_{\text{block}}) + b_m c(t - mT_{\text{block}} - T_m) \right] e^{j k \pi \sin \theta} e^{j 2\pi f_d m T_{\text{block}}} + w_{k,m}(t). \quad (42)$$

**5.2 Angle of Arrival (AoA) Estimation:** At the range-Doppler peak index  $(p_{\text{peak}}, q_{\text{peak}})$ , the complex phase progression across the virtual antennas is represented as a spatial signal vector:

$$\mathbf{s}_{\text{spatial}}[k] = \mathcal{X}_{\text{RD}}[p_{\text{peak}}, k, q_{\text{peak}}] = A' e^{j k \pi \sin \theta}, \quad (43)$$

where  $A'$  denotes the complex amplitude encompassing all processing gains.

To estimate the angle  $\theta$ , we apply a spatial window  $w_{\text{spatial}}[k]$  (such as a Chebyshev or Hamming window to control sidelobes) and compute a spatial DFT of size  $N_{\text{fft, spatial}}$  (often zero-padded to improve angular resolution):

$$\mathcal{X}_{\text{spatial}}[l] = \sum_{k=0}^{N_{\text{virtual}}-1} \mathbf{s}_{\text{spatial}}[k] w_{\text{spatial}}[k] e^{-j \frac{2\pi}{N_{\text{fft, spatial}}} l k}, \quad (44)$$

<sup>2</sup>To better understand why we can set these virtual antennas, see examples in Appendix A.

where  $l$  is the spatial DFT bin index.

This spatial DFT yields a peak at bin  $l_{\text{peak}}$ . By centering the spatial frequencies (e.g., using 'fftshift'), the digital spatial frequency  $\omega_s$  is mapped as:

$$\omega_s = \frac{2\pi(l_{\text{peak}} - N_{\text{fft, spatial}}/2)}{N_{\text{fft, spatial}}} = \pi \sin \theta. \quad (45)$$

From this relationship, the spatial direction sine is:

$$\sin \theta = \frac{2(l_{\text{peak}} - N_{\text{fft, spatial}}/2)}{N_{\text{fft, spatial}}}. \quad (46)$$

Finally, the target azimuth angle  $\theta$  is estimated as:

$$\theta = \arcsin \left( \frac{2(l_{\text{peak}} - N_{\text{fft, spatial}}/2)}{N_{\text{fft, spatial}}} \right). \quad (47)$$

### III. SYSTEM MODEL

#### A. Three-Node System

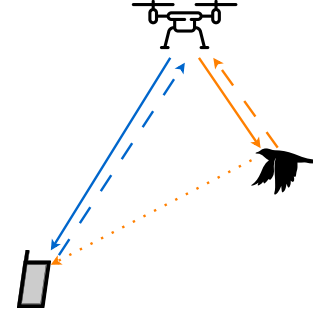


Fig. 6: Three-node system: drone (D), ground user equipment (UE), and moving object (T).

As illustrated in Fig. 6, the three-node system consists of:

- **Drone (D):** A mid-air platform equipped with one transmit and one receive antenna. It acts simultaneously as the Joint Radar-Communication (JRC) transmitter, a communication access point for the UE, and a monostatic radar for sensing the moving object.
- **Ground User Equipment (UE):** A single-antenna mobile receiver that receives the downlink communication signal from the drone, and whose position and velocity can be estimated by the drone.
- **Moving Object (T):** A passive aerial target (e.g., a drone or UAV) that is detected and tracked through radar backscatter.

The drone transmits a PMCW waveform. It employs two quasi-orthogonal Maximum-Length Sequence (MLS) spreading codes. The details are explained in Section III-A2.

1) *Notation and Geometry:* Let  $d_{DU}$ ,  $d_{DT}$  denote the instantaneous Euclidean distances between the drone and the UE, and between the drone and the object, respectively. The corresponding one-way propagation delays are

$$\tau_{DU} = \frac{d_{DU}}{c_0}, \quad (48)$$

$$\tau_{DT} = \frac{d_{DT}}{c_0}, \quad (49)$$



where  $c_0$  is the speed of light. The round-trip monostatic sensing delay for the moving object is

$$\tau_{\text{mono}} = \frac{2d_{DT}}{c_0}. \quad (50)$$

Let  $\mathbf{v}_D$ ,  $\mathbf{v}_U$ , and  $\mathbf{v}_T$  be the velocity vectors of the drone, UE, and moving object, respectively. The radial velocities along each link are projected onto the unit line-of-sight (LOS) vector  $\hat{\mathbf{u}}_{ij}$  from node  $i$  to node  $j$ :

$$v_{r,DU} = (\mathbf{v}_U - \mathbf{v}_D) \cdot \hat{\mathbf{u}}_{DU}, \quad (51)$$

$$v_{r,DT} = (\mathbf{v}_T - \mathbf{v}_D) \cdot \hat{\mathbf{u}}_{DT}. \quad (52)$$

The carrier wavelength for the drone PMCW link is  $\lambda_D = c_0/f_{c,D}$ . The resulting Doppler shifts are

$$f_{D,DU} = -\frac{v_{r,DU}}{\lambda_D}, \quad (53)$$

$$f_{D,DT} = -\frac{v_{r,DT}}{\lambda_D}, \quad (54)$$

and the two-way monostatic Doppler shift for target sensing is  $\nu_{\text{mono}} = 2f_{D,DT}$ .

2) *Transmit Signal*: The drone transmits a Phase-Modulated Continuous Wave (PMCW) waveform. It employs two quasi-orthogonal Maximum-Length Sequence (MLS) spreading codes of length  $N_D$ :  $s_{\text{mls},U}[n] \in \{+1, -1\}$  directed toward the UE, and  $s_{\text{mls},T}[n] \in \{+1, -1\}$  directed toward the moving object. The discrete-time baseband transmit signal during pulse  $m$  (fast-time chip index  $n$ ,  $0 \leq n < N_D$ ,  $0 \leq m < M$ ) is

$$x_D[n, m] = \sqrt{P_D} \left( s_{\text{mls},U}[n] b_m + s_{\text{mls},T}[n] \right), \quad (55)$$

where  $P_D$  is the drone transmit power and  $b_m \in \{+1, -1\}$  is the BPSK communication symbol carried on pulse  $m$ . The two codes are designed to be quasi-orthogonal so that code-domain separation can suppress cross-interference at the respective receivers. The chip duration is  $T_c$ , giving a Pulse Repetition Interval (PRI) of  $T_{\text{PRI}} = N_D T_c$ .

3) *Channel Model*: All drone-to-node and node-to-drone links in the three-node system are SISO channels. Following the generic narrowband link model, the channel coefficient between nodes  $i$  and  $j$  is decomposed as

$$h_{ij} = \sqrt{\beta_{ij}} g_{ij}, \quad (56)$$

where  $\beta_{ij}$  is the large-scale power gain (path loss and shadowing) and  $g_{ij}$  is the complex small-scale fading coefficient. A distance-dependent path-loss model gives

$$\beta_{ij} = C_{ij} \left( \frac{d_{ij}}{d_0} \right)^{-\alpha_{ij}}, \quad (57)$$

with  $C_{ij} = G_i G_j (\lambda_D / (4\pi d_0))^2$  being the reference channel gain at distance  $d_0$ , and  $\alpha_{ij}$  the path-loss exponent.

The time-varying drone-to-UE channel (including the Doppler phase rotation) is modeled as

$$h_{DU}(t) = \sqrt{\beta_{DU}} g_{DU} e^{j2\pi f_{D,DU} t}, \quad (58)$$

and similarly for the drone-to-target and target-to-drone channels:

$$h_{DT}(t) = \sqrt{\beta_{DT}} g_{DT} e^{j2\pi f_{D,DT} t}, \quad (59)$$

$$h_{TD}(t) = \sqrt{\beta_{TD}} g_{TD} e^{j2\pi f_{D,DT} t}. \quad (60)$$

Under channel reciprocity,  $\beta_{TD} = \beta_{DT}$  and  $g_{TD} = g_{DT}$ .

4) *Received Signal at the UE*: The UE receives the drone's downlink PMCW signal:

$$y_U[n, m] = h_{DU} x_D[(n - \ell_{DU})_{N_D}, m] e^{j2\pi f_{D,DU} m T_{\text{PRI}}} + n_U[n, m], \quad (61)$$

where  $\ell_{DU} = \text{round}(\tau_{DU} f_s)$  is the integer chip delay,  $(\cdot)_{N_D}$  denotes modulo- $N_D$ ,  $f_s = 1/T_c$  is the chip-rate sampling frequency, and  $n_U[n, m] \sim \mathcal{CN}(0, \sigma_U^2)$  is additive white Gaussian noise.

Substituting (55) yields the explicit received signal:

$$y_U[n, m] = \sqrt{P_D} h_{DU} \left( s_{\text{mls},U}[(n - \ell_{DU})_{N_D}] b_m + s_{\text{mls},T}[(n - \ell_{DU})_{N_D}] \right) e^{j2\pi f_{D,DU} m T_{\text{PRI}}} + n_U[n, m]. \quad (62)$$

The communication data  $b_m$  is recovered by correlating with the UE code  $s_{\text{mls},U}$ ; the object-sensing code  $s_{\text{mls},T}$  is suppressed by the quasi-orthogonality of the two MLS codes, providing a processing gain of approximately  $N_D$ .

5) *Received Signal at the Drone Receiver (Sensing)*: The drone's receive chain collects the backscatter from both the UE (cooperative echo / passive reflection) and the moving object. In the general case, the composite received signal at the drone is

$$y_D[n, m] = y_{D,UE}[n, m] + y_{D,T}[n, m] + n_D[n, m], \quad (63)$$

where  $n_D[n, m] \sim \mathcal{CN}(0, \sigma_D^2)$  is receiver noise.

a) *UE-Related Echo*: The echo returning from the UE path ( $D \rightarrow U \rightarrow D$ ) is modeled as

$$y_{D,UE}[n, m] = \eta_U \beta_{DU} g_{DU}^2 \sqrt{P_D} s_{\text{mls},U}[(n - \ell_{2DU})_{N_D}] \times b_m e^{j2\pi \nu_{DU} m T_{\text{PRI}}}, \quad (64)$$

where  $\eta_U$  is the UE reflection coefficient,  $\ell_{2DU} = \text{round}(2\tau_{DU} f_s)$  is the round-trip chip delay, and  $\nu_{DU} = 2f_{D,DU}$  is the round-trip Doppler shift.

b) *Moving-Object Echo*: The monostatic target echo from the object ( $D \rightarrow T \rightarrow D$ ) is modeled as

$$y_{D,T}[n, m] = \eta_T \beta_{DT} g_{DT}^2 \sqrt{P_D} s_{\text{mls},T}[(n - \ell_{\text{mono}})_{N_D}] \times e^{j2\pi \nu_{\text{mono}} m T_{\text{PRI}}}, \quad (65)$$

where  $\eta_T$  is the complex target reflection coefficient (related to the Radar Cross Section,  $\sigma_T$ ),  $\ell_{\text{mono}} = \text{round}(\tau_{\text{mono}} f_s)$ , and  $\nu_{\text{mono}} = 2f_{D,DT}$  is the round-trip Doppler. Under channel reciprocity,  $\beta_{TD} = \beta_{DT}$  and  $g_{TD} = g_{DT}$ .

The full monostatic received signal at the drone is therefore:

$$\begin{aligned}
 y_D[n, m] &= \underbrace{\eta_T \beta_{DT} g_{DT}^2 \sqrt{P_D} s_{\text{mls}, T}[(n - \ell_{\text{mono}})N_D]}_{\text{object sensing echo}} \times e^{j2\pi\nu_{\text{mono}} m T_{\text{PRI}}} \\
 &+ \underbrace{\eta_U \beta_{DU} g_{DU}^2 \sqrt{P_D} s_{\text{mls}, U}[(n - \ell_{2DU})N_D]}_{\text{UE-related echo (interference for object sensing)}} \times b_m e^{j2\pi\nu_{DU} m T_{\text{PRI}}} \\
 &+ n_D[n, m]. \tag{66}
 \end{aligned}$$

The object sensing echo is isolated by correlating  $y_D[n, m]$  with  $s_{\text{mls}, T}$ ; the UE echo term is suppressed by code quasi-orthogonality with gain  $N_D$ .

### B. Four-Node System

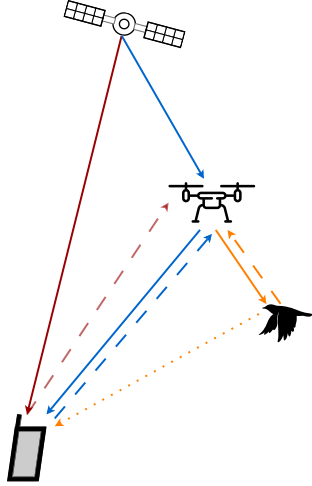


Fig. 7: Four-node system: LEO satellite (S), mid-air radar/drone (R), ground user equipment (U), and moving target (T).

The four-node system extends the three-node architecture by introducing a **LEO Satellite (S)** as the primary downlink communication transmitter and an additional bistatic radar illuminator. The four nodes (Fig. 7) are:

- **Satellite (S):** A multi-antenna LEO satellite with  $N_s$  transmit elements, serving as the downlink communication source for the UE and as a bistatic radar illuminator for the moving target.
- **Mid-Air Radar / Drone (R):** A single transmit/receive antenna aerial platform performing monostatic radar sensing and receiving bistatic reflections.
- **Ground UE (U):** A single-antenna mobile receiver receiving the satellite downlink subject to radar interference.

- **Moving Target (T):** An aerial object with radar cross section  $\sigma_T$ , sensed by both monostatic ( $R \rightarrow T \rightarrow R$ ) and bistatic ( $S \rightarrow T \rightarrow R$ ) paths.

1) *Signal Separation Strategy:* Two waveform families and three orthogonality mechanisms are combined to separate the signals in (98) and (91):

- **Satellite links ( $S \rightarrow U, S \rightarrow T, S \rightarrow R$ ):** OTFS waveforms are adopted to robustly handle the large Doppler and delay spreads of LEO satellite channels.
- **Local drone links ( $R \rightarrow T \rightarrow R$ ):** PMCW waveforms with MLS code  $s_{\text{mls}, R}$  are used for high-resolution monostatic sensing.
- **Code-domain orthogonality:** The quasi-orthogonality of  $s_{\text{mls}, S}$  and  $s_{\text{mls}, R}$  suppresses cross-code leakage with a processing gain of  $\min(N_S, N_R)$ .
- **Time-domain orthogonality:** Dedicated PMCW slots are allocated to avoid overlap between satellite and radar transmissions.
- **Frequency-domain orthogonality:** Distinct subbands are assigned to the satellite and radar links where code-domain separation alone is insufficient.

The direct-path interference  $y_{SR}$  in (98) appears at delay bin  $\ell_{SR}$  and Doppler bin  $f_{D, SR}$ , which are generally well separated from the target echo located at  $\ell_{bi}$  and  $\nu_{bi}$ . Residual direct-path interference is mitigated by spatial nulling via the satellite beamforming vector  $\mathbf{w}_S$  or by software cancellation in the bistatic processing branch.

2) *Notation and Geometry:* The instantaneous inter-node Euclidean distances are:

$$\begin{aligned}
 d_{SU}(t) &= \|\mathbf{p}_S(t) - \mathbf{p}_U(t)\|, \\
 d_{SR}(t) &= \|\mathbf{p}_S(t) - \mathbf{p}_R(t)\|, \\
 d_{RU}(t) &= \|\mathbf{p}_R(t) - \mathbf{p}_U(t)\|, \\
 d_{ST}(t) &= \|\mathbf{p}_S(t) - \mathbf{p}_T(t)\|, \\
 d_{RT}(t) &= \|\mathbf{p}_R(t) - \mathbf{p}_T(t)\|, \tag{67}
 \end{aligned}$$

where  $\mathbf{p}_i(t) \in \mathbb{R}^3$  is the position vector of node  $i$  at time  $t$ .

The propagation delays for all active paths are:

$$\tau_{SU}(t) = d_{SU}(t)/c_0, \tag{68}$$

$$\tau_{SR}(t) = d_{SR}(t)/c_0, \tag{69}$$

$$\tau_{RU}(t) = d_{RU}(t)/c_0, \tag{70}$$

$$\tau_{\text{mono}}(t) = 2d_{RT}(t)/c_0, \tag{71}$$

$$\tau_{bi}(t) = (d_{ST}(t) + d_{RT}(t))/c_0. \tag{72}$$

The Doppler shifts on each propagation link are obtained

from the radial velocity projections:

$$f_{D,SU}(t) = -\frac{(\mathbf{v}_U - \mathbf{v}_S) \cdot \hat{\mathbf{u}}_{SU}}{\lambda_S}, \quad (73)$$

$$f_{D,SR}(t) = -\frac{(\mathbf{v}_R - \mathbf{v}_S) \cdot \hat{\mathbf{u}}_{SR}}{\lambda_S}, \quad (74)$$

$$f_{D,RU}(t) = -\frac{(\mathbf{v}_U - \mathbf{v}_R) \cdot \hat{\mathbf{u}}_{RU}}{\lambda_R}, \quad (75)$$

$$f_{D,ST}(t) = -\frac{(\mathbf{v}_T - \mathbf{v}_S) \cdot \hat{\mathbf{u}}_{ST}}{\lambda_S}, \quad (76)$$

$$f_{D,RT}(t) = -\frac{(\mathbf{v}_T - \mathbf{v}_R) \cdot \hat{\mathbf{u}}_{RT}}{\lambda_R}, \quad (77)$$

where  $\lambda_S = c_0/f_{c,S}$  and  $\lambda_R = c_0/f_{c,R}$  are the satellite and radar carrier wavelengths, respectively. The cumulative two-way Doppler shifts for the sensing paths are

$$\nu_{\text{mono}}(t) = 2 f_{D,RT}(t), \quad (78)$$

$$\nu_{\text{bi}}(t) = f_{D,ST}(t) + f_{D,RT}(t). \quad (79)$$

3) *Satellite Antenna Model:* The satellite is equipped with a Uniform Linear Array (ULA) of  $N_s$  elements with half-wavelength spacing  $d = \lambda_S/2$ . The transmit spatial steering vector toward direction  $\vartheta$  is

$$\mathbf{a}_{N_s}(\vartheta) = [1, e^{j\pi \sin \vartheta}, \dots, e^{j\pi(N_s-1) \sin \vartheta}]^T \in \mathbb{C}^{N_s \times 1}. \quad (80)$$

A transmit beamforming vector  $\mathbf{w}_S \in \mathbb{C}^{N_s \times 1}$  (with  $\|\mathbf{w}_S\| = 1$ ) is applied to focus energy toward the UE and simultaneously illuminate the target.

4) *Transmit Signals:*

a) *Satellite Transmit Signal:* The satellite modulates a BPSK communication symbol sequence  $\{b_m\}$ ,  $b_m \in \{+1, -1\}$ , onto a length- $N_S$  MLS spreading code  $s_{\text{mls},S}[n] \in \{+1, -1\}$ , and applies spatial beamforming. The discrete-time baseband transmit signal vector is

$$\mathbf{x}_S[n, m] = \sqrt{P_S} \mathbf{w}_S s_{\text{mls},S}[n] b_m \in \mathbb{C}^{N_s \times 1}, \quad (81)$$

where  $P_S$  is the satellite transmit power and the PRI is  $T_{\text{PRI},S} = N_S T_c$ .

b) *Radar Transmit Signal:* The radar transmits a dedicated monostatic sensing waveform using a distinct quasi-orthogonal MLS code  $s_{\text{mls},R}[n] \in \{+1, -1\}$  of length  $N_R$ :

$$x_R[n, m] = \sqrt{P_R} s_{\text{mls},R}[n], \quad (82)$$

where  $P_R$  is the radar transmit power and the PRI is  $T_{\text{PRI},R} = N_R T_c$ .

5) *Channel Models:*

a) *Space-to-Ground/Mid-Air Channels (MISO Rician):* Due to the high LEO altitude, all satellite links ( $S \rightarrow U$ ,  $S \rightarrow T$ ,  $S \rightarrow R$ ) experience a high Rician  $K$ -factor. The MISO channel row vector  $\mathbf{h}_{Sj}^H \in \mathbb{C}^{1 \times N_s}$  is modeled as

$$\mathbf{h}_{Sj}^H = \sqrt{\beta_{Sj}} e^{j\phi_{Sj}} e^{j2\pi f_{D,Sj} t} \left( \sqrt{\frac{K_{Sj}}{K_{Sj}+1}} \mathbf{a}_{N_s}^H(\vartheta_S^j) + \sqrt{\frac{1}{K_{Sj}+1}} \mathbf{g}_{Sj}^H \right), \quad (83)$$

where  $\beta_{Sj}$  is the large-scale path gain of the  $S \rightarrow j$  link,  $\phi_{Sj}$  is a random initial phase,  $K_{Sj}$  is the Rician factor,  $\vartheta_S^j$  is

the Angle of Departure (AoD) from the satellite array toward node  $j$ , and  $\mathbf{g}_{Sj}^H \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_{N_s})$  represents the scattered NLoS components. Specifically:

$$\mathbf{h}_{SU}^H: \text{satellite-to-UE channel}, \quad (84)$$

$$\mathbf{h}_{ST}^H: \text{satellite-to-target channel}, \quad (85)$$

$$\mathbf{h}_{SR}^H: \text{satellite-to-radar direct channel}. \quad (86)$$

b) *Aerial/Terrestrial Channels (SISO):* All links between mid-air and ground nodes ( $R \rightarrow U$ ,  $R \rightarrow T$ ,  $T \rightarrow R$ ) are SISO channels:

$$h_{RU}(t) = \sqrt{\beta_{RU}} g_{RU} e^{j2\pi f_{D,RU} t}, \quad (87)$$

$$h_{RT}(t) = \sqrt{\beta_{RT}} g_{RT} e^{j2\pi f_{D,RT} t}, \quad (88)$$

$$h_{TR}(t) = \sqrt{\beta_{TR}} g_{TR} e^{j2\pi f_{D,RT} t}. \quad (89)$$

Under channel reciprocity,  $\beta_{TR} = \beta_{RT}$  and  $g_{TR} = g_{RT}$ .

6) *Received Signal at the UE:* The ground UE receives the superposition of the satellite communication signal and the co-channel radar sensing interference:

$$\begin{aligned} y_U[n, m] = & \underbrace{\mathbf{h}_{SU}^H \mathbf{x}_S[n - \ell_{SU}, m] e^{j2\pi f_{D,SU} m T_{\text{PRI},S}}}_{\text{desired satellite signal}} \\ & + \underbrace{h_{RU} x_R[n - \ell_{RU}, m] e^{j2\pi f_{D,RU} m T_{\text{PRI},R}}}_{\text{radar interference}} \\ & + n_U[n, m], \end{aligned} \quad (90)$$

where  $\ell_{SU} = \text{round}(\tau_{SU} f_s)$ ,  $\ell_{RU} = \text{round}(\tau_{RU} f_s)$ , and  $n_U[n, m] \sim \mathcal{CN}(0, \sigma_U^2)$  is the UE receiver noise.

Substituting (81) and (82) yields:

$$\begin{aligned} y_U[n, m] = & \sqrt{P_S} (\mathbf{h}_{SU}^H \mathbf{w}_S) s_{\text{mls},S}[(n - \ell_{SU})_{N_S}] b_m \\ & \times e^{j2\pi f_{D,SU} m T_{\text{PRI},S}} \\ & + \sqrt{P_R} h_{RU} s_{\text{mls},R}[(n - \ell_{RU})_{N_R}] \\ & \times e^{j2\pi f_{D,RU} m T_{\text{PRI},R}} \\ & + n_U[n, m]. \end{aligned} \quad (91)$$

The effective complex channel coefficient for the communication link is  $h_{\text{comm}} = \sqrt{P_S} (\mathbf{h}_{SU}^H \mathbf{w}_S)$ . Note that despreading by correlating  $y_U[n, m]$  with  $s_{\text{mls},S}$  boosts the desired signal by a factor of  $N_S$  while suppressing the radar interference, since the two MLS codes are quasi-orthogonal.

7) *Received Signal at the Radar Receiver:* The radar receiver collects the composite signal containing three contributions: the monostatic target echo, the bistatic target echo reflected from the satellite illumination, and the direct-path satellite interference:

$$y_R[n, m] = y_{RTR}[n, m] + y_{STR}[n, m] + y_{SR}[n, m] + n_R[n, m], \quad (92)$$

where  $n_R[n, m] \sim \mathcal{CN}(0, \sigma_R^2)$  is the radar receiver noise.

a) *Monostatic Echo ( $R \rightarrow T \rightarrow R$ ):* The radar's own waveform is reflected from the moving target back to the radar. Under channel reciprocity ( $\beta_{TR} = \beta_{RT}$ ,  $g_{TR} = g_{RT}$ ):

$$\begin{aligned} y_{RTR}[n, m] = & \eta_T \beta_{RT} g_{RT}^2 \sqrt{P_R} s_{\text{mls},R}[(n - \ell_{\text{mono}})_{N_R}] \\ & \times e^{j2\pi \nu_{\text{mono}} m T_{\text{PRI},R}}, \end{aligned} \quad (93)$$

where  $\ell_{\text{mono}} = \text{round}(\tau_{\text{mono}} f_s)$  and  $\nu_{\text{mono}}$  is given by (78).



b) *Bistatic Echo* ( $S \rightarrow T \rightarrow R$ ): The satellite signal illuminates the target; the reflected echo is collected by the radar:

$$y_{STR}[n, m] = \eta_T \sqrt{\beta_{ST} \beta_{TR}} g_{ST} g_{TR} \sqrt{P_S} (\mathbf{h}_{ST}^H \mathbf{w}_S) \times s_{\text{mIs}, S}[(n - \ell_{\text{bi}})_{N_S}] b_m e^{j2\pi \nu_{\text{bi}} m T_{\text{PRI}, S}}, \quad (94)$$

where  $\ell_{\text{bi}} = \text{round}(\tau_{\text{bi}} f_s)$  and  $\nu_{\text{bi}}$  is given by (79).

c) *Direct-Path Satellite Signal (Interference)*: The satellite's downlink transmission is received directly by the radar, constituting strong direct-path interference:

$$y_{SR}[n, m] = \sqrt{P_S} (\mathbf{h}_{SR}^H \mathbf{w}_S) s_{\text{mIs}, S}[(n - \ell_{SR})_{N_S}] \times b_m e^{j2\pi f_{D, SR} m T_{\text{PRI}, S}}, \quad (95)$$

where  $\ell_{SR} = \text{round}(\tau_{SR} f_s)$ .

Combining the three contributions, the complete received signal at the radar is:

$$y_R[n, m] = \eta_T \beta_{RT} g_{RT}^2 \sqrt{P_R} s_{\text{mIs}, R}[(n - \ell_{\text{mono}})_{N_R}] \quad (96)$$

$$\begin{aligned} & \underbrace{\times e^{j2\pi \nu_{\text{mono}} m T_{\text{PRI}, R}}}_{\text{monostatic target echo}} \\ & + \eta_T \sqrt{\beta_{ST} \beta_{TR}} g_{ST} g_{TR} \sqrt{P_S} (\mathbf{h}_{ST}^H \mathbf{w}_S) b_m \\ & \underbrace{\times s_{\text{mIs}, S}[(n - \ell_{\text{bi}})_{N_S}] e^{j2\pi \nu_{\text{bi}} m T_{\text{PRI}, S}}}_{\text{bistatic target echo}} \\ & + \sqrt{P_S} (\mathbf{h}_{SR}^H \mathbf{w}_S) s_{\text{mIs}, S}[(n - \ell_{SR})_{N_S}] b_m \end{aligned} \quad (97)$$

$$\begin{aligned} & \underbrace{\times e^{j2\pi f_{D, SR} m T_{\text{PRI}, S}}}_{\text{direct-path satellite interference}} \\ & + n_R[n, m]. \end{aligned} \quad (98)$$

#### IV. PROPOSED PROBLEM

Pre-Compensation at the BS: Since the BS estimates the radar parameters ( $R, v$ ), it can calculate the expected downlink Doppler shift ( $f_{d, \text{comm}} = \frac{1}{2} f_{d, \text{radar}}$ ) and propagation delay.

- The BS **pre-compensates** the communication signal before transmission. It applies a frequency shift of  $-f_{d, \text{comm}}$  and a time-delay offset to the downlink transmit waveform.
- **Result at UE:** When the signal arrives at the UE, the Doppler shift and delay offsets are already cancelled out. The UE receives a “Doppler-free” and “delay-aligned” signal, meaning the UE does not need to perform complex channel tracking or estimation for these parameters.
- **Prediction Location: At the BS.** The BS predicts the UE's trajectory to apply the correct pre-compensation for the next block.

#### APPENDIX

##### MIMO Radar: Why Spatial FFT Extracts the Angle of Arrival?

This section explains the physical and mathematical principles underlying how a Spatial Fast Fourier Transform (FFT) applied across the virtual antenna array elements extracts a target's physical azimuth angle of arrival (AoA), denoted by  $\theta$ .

##### A. Physical to Virtual Array Mapping

In colocated MIMO radar systems, a thinned physical antenna array is typically utilized to synthesize a dense virtual array, significantly increasing the spatial resolution without requiring a large number of physical transceivers. For example, consider the following configuration:

- **Transmit (Tx) Array:** Consists of  $N_{\text{Tx}} = 4$  elements with a spacing of  $d_{\text{Tx}} = \frac{\lambda}{2}$ . The physical positions of the Tx elements are given by:

$$x_{\text{Tx}}[u] = u \cdot \frac{\lambda}{2}, \quad u \in \{0, 1, 2, 3\}. \quad (99)$$

- **Receive (Rx) Array:** Consists of  $N_{\text{Rx}} = 4$  elements with a spacing of  $d_{\text{Rx}} = 4 \cdot \frac{\lambda}{2}$ . The physical positions of the Rx elements are given by:

$$x_{\text{Rx}}[v_r] = v_r \cdot 4 \cdot \frac{\lambda}{2}, \quad v_r \in \{0, 1, 2, 3\}. \quad (100)$$

The synthesized virtual antenna positions are defined by the spatial convolution (algebraic sum) of the Tx and Rx positions:

$$d_{\text{virtual}}(u, v_r) = x_{\text{Tx}}[u] + x_{\text{Rx}}[v_r]. \quad (101)$$

By mapping all  $N_{\text{Tx}} N_{\text{Rx}} = 16$  combinations ( $u \in \{0, 1, 2, 3\}$  and  $v_r \in \{0, 1, 2, 3\}$ ) into a sorted one-dimensional vector, we synthesize a filled Uniform Linear Array (ULA) of  $N_{\text{virtual}} = 16$  elements spaced sequentially at:

$$x_{\text{virtual}}[k] = k \cdot \frac{\lambda}{2}, \quad k \in \{0, 1, \dots, 15\}. \quad (102)$$

Stacking the range-Doppler maps corresponding to these 16 virtual elements yields the third dimension (the spatial channel dimension) of the radar data cube. Along this spatial axis, the complex phase of the target signal shifts progressively from one virtual element to the next. This constant phase increment  $\Delta\phi$  is dictated by the propagation path length difference  $\Delta R$  between adjacent virtual elements:

$$\Delta R = \frac{\lambda}{2} \sin \theta, \quad (103)$$

$$\Delta\phi = \frac{2\pi}{\lambda} \Delta R = \pi \sin \theta. \quad (104)$$

Consequently, for a virtual ULA of  $N_{\text{virtual}}$  elements, the received spatial signal vector  $\mathbf{x}[k]$  (extracted at the target's range-Doppler peak index) is modeled as:

$$\mathbf{x}[k] = A \cdot e^{jk\Delta\phi} = A \cdot e^{jk\pi \sin \theta}, \quad (105)$$

where  $k \in \{0, 1, \dots, N_{\text{virtual}} - 1\}$  is the virtual antenna index, and  $A$  is the complex amplitude incorporating the fast-time matched filtering and slow-time Doppler processing gains.

### B. Spatial Frequency Mapping via FFT

The Discrete Fourier Transform (DFT), commonly implemented via the FFT algorithm, decomposes a discrete sequence into its constituent frequency components. When applied to the spatial vector  $\mathbf{x}[k]$ , the FFT treats the virtual antenna index  $k$  as the independent variable (analogous to time in temporal signal processing) and the phase progression  $\Delta\phi$  as the spatial frequency.

Thus, the spatial FFT transforms the signal from the spatial antenna domain to the spatial frequency (angular) domain. The output spectrum exhibits a sharp peak at the normalized digital spatial frequency:

$$f_{\text{peak}} = \frac{\omega_s}{2\pi} = \frac{\pi \sin \theta}{2\pi} = \frac{\sin \theta}{2}, \quad (106)$$

where  $f_{\text{peak}} \in [-0.5, 0.5)$  represents the centered normalized frequency. By identifying the peak bin index, the physical azimuth angle of arrival  $\theta$  is directly resolved as:

$$\sin \theta = 2 \cdot f_{\text{peak}}, \quad (107)$$

$$\theta = \arcsin(2 \cdot f_{\text{peak}}). \quad (108)$$

### C. Case Studies and Examples

1) *Example 1: Transmit and Receive Antenna Alignment:* Consider the physical alignment of transmit (Tx) and receive (Rx) antennas on the radar device, as illustrated in Fig. 8. The path length difference is defined by  $\Delta R = \frac{\lambda}{2} \sin \theta$  and the corresponding phase shift is  $\Delta\phi = \frac{2\pi}{\lambda} \Delta R$ . We trace the path differences relative to a reference element pair:

- **Reference Path ( $\mathbf{Tx}_1 \rightarrow \mathbf{Rx}_1$ ):** Path difference is  $0 \cdot \frac{\lambda}{2} \sin \theta = 0\Delta R$ , establishing a reference phase of  $0\Delta\phi$ .
- **Path  $\mathbf{Tx}_2 \rightarrow \mathbf{Rx}_1$ :** Since  $\mathbf{Tx}_2$  is physically displaced from  $\mathbf{Tx}_1$  by  $\frac{\lambda}{2}$ , the incremental distance is  $1 \cdot \frac{\lambda}{2} \sin \theta = 1\Delta R$ , yielding a phase difference of  $1\Delta\phi$ .
- **Path  $\mathbf{Tx}_3 \rightarrow \mathbf{Rx}_1$ :** Path length increment is  $2\Delta R$ , yielding a phase difference of  $2\Delta\phi$ .
- **Path  $\mathbf{Tx}_4 \rightarrow \mathbf{Rx}_1$ :** Path length increment is  $3\Delta R$ , yielding a phase difference of  $3\Delta\phi$ .
- **Path  $\mathbf{Tx}_1 \rightarrow \mathbf{Rx}_2$ :** Since  $\mathbf{Rx}_2$  is physically displaced from  $\mathbf{Rx}_1$  by  $4 \cdot \frac{\lambda}{2}$ , the incremental distance is  $4 \cdot \frac{\lambda}{2} \sin \theta = 4\Delta R$ , yielding a phase difference of  $4\Delta\phi$ .
- ...

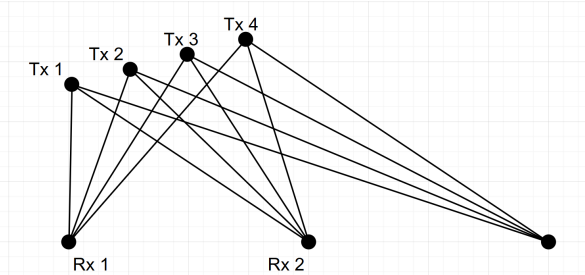


Fig. 8: Antenna element alignments and phase progression for Tx and Rx arrays.

Note that in a direct line-of-sight (LoS) setup between two multi-antenna JRC transceivers that are angularly displaced (i.e., not facing each other directly), the departure angle  $\theta_1$  from the transmitter and the arrival angle  $\theta_2$  at the receiver exhibit a symmetric geometric relationship satisfying  $\sin \theta_1 = \sin \theta_2$ .

2) *Example 2: Phase Progression with Scattering from a Target:* Consider a scenario where the transmitted signal propagates from the transmitter array to a target (scatterer)  $U$ , and then reflects toward the receiver array, as illustrated in Fig. 9. Let  $\theta_1$  represent the angle of departure (AoD) from the transmitter array to the target, and  $\theta_2$  represent the angle of arrival (AoA) from the target to the receiver array.

- **Reference Path ( $\mathbf{T}_1 \rightarrow U \rightarrow \mathbf{R}_1$ ):** Serving as the baseline, the path difference is set to  $0 \cdot \frac{\lambda}{2} \sin \theta_1 = 0\Delta R_1$ , with reference phase  $0\Delta\phi_1$ .
- **Path  $\mathbf{T}_2 \rightarrow U \rightarrow \mathbf{R}_1$ :** The incremental distance is solely due to the displacement between  $\mathbf{T}_2$  and  $\mathbf{T}_1$  relative to  $U$ . Since they are separated by  $\frac{\lambda}{2}$ , the path increment is  $1 \cdot \frac{\lambda}{2} \sin \theta_1 = 1\Delta R_1$ , and the phase difference is  $1\Delta\phi_1$ .
- **Path  $\mathbf{T}_3 \rightarrow U \rightarrow \mathbf{R}_1$ :** Path length increment is  $2\Delta R_1$ , yielding phase difference  $2\Delta\phi_1$ .
- **Path  $\mathbf{T}_4 \rightarrow U \rightarrow \mathbf{R}_1$ :** Path length increment is  $3\Delta R_1$ , yielding phase difference  $3\Delta\phi_1$ .
- **Path  $\mathbf{T}_1 \rightarrow U \rightarrow \mathbf{R}_2$ :** The incremental distance is determined by the displacement of the receiver elements. Since  $\mathbf{R}_2$  is spaced  $4 \cdot \frac{\lambda}{2}$  away from  $\mathbf{R}_1$ , the path length increment is  $4 \cdot \frac{\lambda}{2} \sin \theta_2 = 4\Delta R_2$ , yielding a phase shift of  $4\Delta\phi_2$ .
- ...

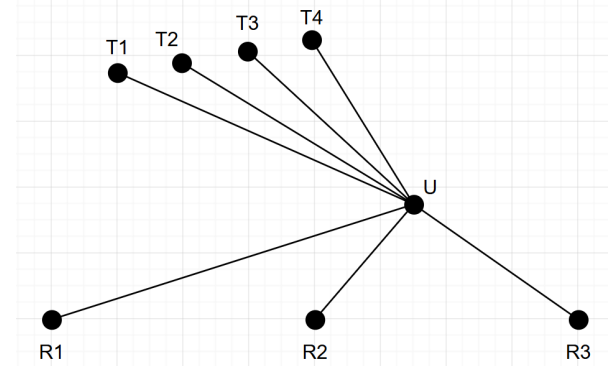


Fig. 9: Scattering geometry and phase progression via target reflection.

To maintain a uniform phase progression across the virtual array (i.e., ensuring  $\Delta R_1 = \Delta R_2$  and  $\Delta\phi_1 = \Delta\phi_2$ ), the condition  $\theta_1 = \theta_2$  must be satisfied. This implies that the transmit and receive antenna arrays must be “structural partners”—they must lie on the same baseline, face the same direction, and share an identical geometric definition of the azimuth angle  $\theta$ .

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